

Neel Maddu

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Summary

I've built AI-powered tools like a multimodal search engine using CLIP, FAISS, and PyTorch, and led a team of 8 to ship production-level iOS apps with Core ML at Georgia Tech. I turn ambiguous problems into documented, repeatable systems — from architecting MVVM codebases to reviewing 40+ PRs and mentoring junior developers. I'm looking to apply my full-stack and ML engineering skills to build impactful software at scale.

EDUCATION

Georgia Institute of Technology

Bachelor of Science, Computer Science

May 2027

Atlanta, GA

• **GPA:** 4.0

• **Coursework:** Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Objects & Design, Computer Organization & Programming, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Probability & Statistics

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, FAISS
- **Data:** pandas, NumPy
- **iOS Development:** Swift, SwiftUI, Core ML, MVVM
- **Backend & Databases:** SQL, MySQL, MongoDB, Firebase, Node.js
- **Programming Languages:** Python, Java, C, JavaScript, C++, Next.js, React
- **Tools:** Git, GitHub, CI/CD, Jenkins, GitHub Actions, Docker, CMake
- **Cloud:** AWS, Azure

PROJECTS

Multimodal AI Search Engine

Feb 2025 - Nov 2025

- Developed a visual search engine by indexing over 10,000 images using CLIP embeddings and FAISS, retrieving semantically relevant images from natural language prompts with an average query time of 2 seconds.
- Built and deployed a Streamlit interface for real-time, scrollable image results, demonstrating practical ML deployment and cross-modal embedding techniques.

JotDown

Sep 2025 - Dec 2025

- Architected the application's UI with SwiftUI, employing an MVVM design pattern to build a responsive and minimalist interface for frictionless data entry.
- Integrated on-device Core ML models to perform Natural Language Processing, enabling a semantic search feature that recognizes entities like dates and keywords within user queries.
- Designed a data visualization module to generate a knowledge graph from user notes, modeling semantic relationships to uncover latent themes and connections.

Real-Time ASL Interpreter

May 2025 - Present

- Engineered a real-time ASL interpreter for static alphabet letters using MediaPipe and scikit-learn, achieving 95% classification accuracy on gestures captured from a live webcam feed.
- Integrating text-to-speech functionality that activates on new predictions, with a mute toggle and a "None" class for unclear gestures for a better user experience.

EXPERIENCE

Georgia Tech iOS Club

Jan 2026 - Present

Senior iOS Developer

Atlanta, GA

- Led a team of 8 developers to build iOS apps with Swift, Core ML, and Firebase, delivering 3 club projects used by 150+ members with an average 4.7/5 satisfaction rating
- Collaborated with designers and developers to architect SwiftUI features using MVVM, reducing onboarding drop-off by 30% and increasing average session duration by 2 minutes
- Reviewed 40+ pull requests in GitHub, provide technical feedback using code-review guidelines, and mentor 4 junior developers, reducing bug rate by 25% and cutting feature delivery time by 1 week
- Contribute to production-level iOS codebases by writing new features, fixing bugs, and adding unit tests, achieving 99% test coverage and zero rollback releases across 3 release cycles

Waffle House

May 2024 - Feb 2025

Salesperson

Cumming, GA

- Delivered customer service in a high-volume, fast-paced environment, using the POS system to process an average of 120 orders per shift and ensuring 99% order accuracy, which boosted customer satisfaction scores
- Managed point-of-sale transactions using the restaurant POS system and accurately balanced cash and tip drawers each shift, maintaining zero cash discrepancies
- Collaborated with team members using handheld radios and a shared task checklist to keep service efficient and resolve issues quickly during peak hours, resulting in smoother service flow and shorter customer wait times